Vector Calculus

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1 Euclidean space

1.1 Basic Concepts

We define the **Euclidean space** $\mathbb{R}^n$, $n \geq 1$, as the set of all elements in $\mathbb{R}$:

$$x = (x_1, x_2, \ldots, x_n) \quad \text{where } x_i \in \mathbb{R} \text{ for } i = 1, 2, \ldots, n.$$

The **Standard basis** of $\mathbb{R}^n$ is given by the vectors:

$$e_1 = (1, 0, \ldots, 0), \quad e_2 = (0, 1, \ldots, 0), \quad \ldots, \quad e_n = (0, 0, \ldots, 1).$$

Then, any vector $x \in \mathbb{R}^n$ can be expressed as a linear combination of the standard basis vectors:

$$x = x_1 e_1 + x_2 e_2 + \ldots + x_n e_n.$$

1.2 Properties of Euclidean Space

The Euclidean space $\mathbb{R}^n$ has the following properties:

- **Addition:** For any $x, y \in \mathbb{R}^n$, the sum $x + y$ is defined component-wise:

$$x + y = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n).$$

- **Scalar Multiplication:** For any scalar $\alpha \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$, the product αx is defined as:

$$\alpha x = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n).$$

- **Scalar Distributivity:** For any scalars $\alpha, \beta \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$:

$$(\alpha + \beta)x = \alpha x + \beta x.$$

- **Vector Distributivity:** For any scalar $\alpha \in \mathbb{R}$ and vectors $x, y \in \mathbb{R}^n$:

$$\alpha(x + y) = \alpha x + \alpha y.$$

- **Scalar Commutativity:** For any scalars $\alpha, \beta \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$:

$$\alpha(\beta x) = (\alpha\beta)x.$$

- **Multiplicative Identity:** For any vector $x \in \mathbb{R}^n$:

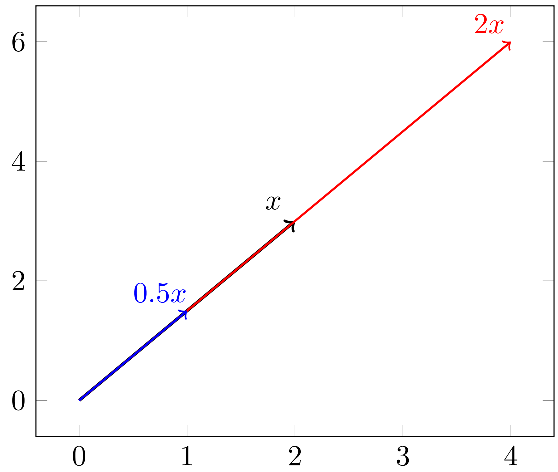
$$1 \cdot x = x.$$

- **Multiplicative Zero:** For any vector $x \in \mathbb{R}^n$:

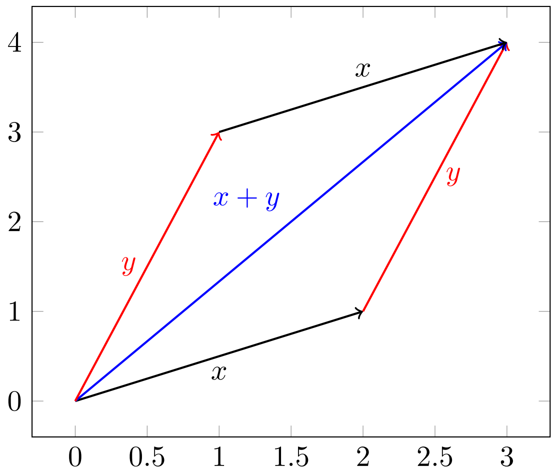
$$0 \cdot x = 0.$$

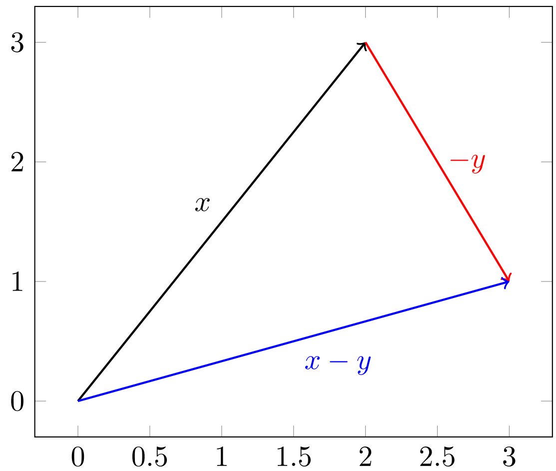
1.3 Geometric Interpretation

The Euclidean space $\mathbb{R}^n$ can be visualized as an n-dimensional space where each point corresponds to a vector. Given a vector $x = (x_1, x_2, \ldots, x_n)$, and a scalar $\alpha \in \mathbb{R}$, the vector αx represents a point that is scaled by the factor α in the direction of x:



Given two vectors $x, y \in \mathbb{R}^n$, their sum $x + y$ represents the vector obtained by placing the tail of y at the head of x. Similarly, the difference $x - y$ can be visualized by reversing the direction of y and adding it to x.





1.4 Norm and Distance

The Euclidian space is a **normed vector space** with the norm defined as:

$$\begin{aligned} \|\cdot\| :\ & \mathbb{R}^n \to \mathbb{R} \\ & x \mapsto \|x\| = \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}. \end{aligned}$$

The norm satisfies the following properties for all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- **Non-negativity:** $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$.
- **Scalar Multiplication:** $\|\alpha x\| = |\alpha| \|x\|$.
- **Triangle Inequality:** $\|x + y\| \leq \|x\| + \|y\|$.

Remark

Associated with the norm is the **distance function** $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defined by:

$$d(x, y) = \|x - y\|.$$

The distance function satisfies the following properties for all $x, y, z \in \mathbb{R}^n$:

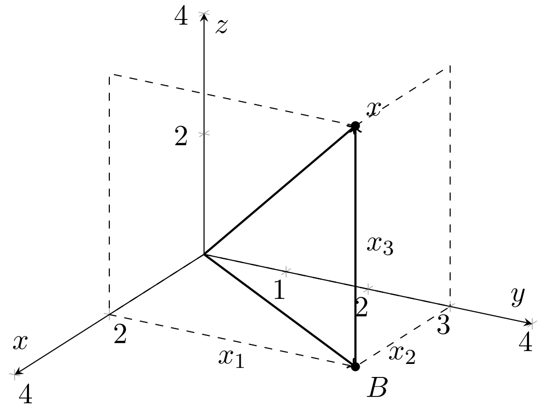
- **Non-negativity:** $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$.
- **Symmetry:** $d(x, y) = d(y, x)$.
- **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

1.4.1 Norms in $\mathbb{R}^3$

In $\mathbb{R}^3$, the norm of a vector $x = (x_1, x_2, x_3)$ is given by:

$$\|x\| = d(O, x) = OB^2 + x_3^2 = \sqrt{x_1^2 + x_2^2 + x_3^2},$$

where O is the origin and B is the projection of x onto the xy-plane:



1.5 Inner or Scalar Product

Let $x, y \in \mathbb{R}^n$. The **inner product** (or **scalar product**) of x and y is defined as:

$$x \cdot y = \langle x, y \rangle = x_1y_1 + x_2y_2 + \ldots + x_ny_n.$$

The inner product satisfies the following properties for all $x, y, z \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- **Positive Definite:** $x \cdot x \geq 0$ and $x \cdot x = 0$ if and only if $x = 0$.
- **Symmetry:** $x \cdot y = y \cdot x$.
- **Bilinear Form:** $\langle \lambda x + \mu y, z \rangle = \lambda \langle x, z \rangle + \mu \langle y, z \rangle$ for all $\lambda, \mu \in \mathbb{R}$.

1.6 Cauchy-Schwarz Inequality

For any vectors $x, y \in \mathbb{R}^n$, the Cauchy-Schwarz inequality states that:

$$|x \cdot y| \leq \|x\|\|y\|.$$

Equality holds if and only if x and y are linearly dependent.

Proof. If $y = \lambda x$ for some $\lambda \in \mathbb{R}$, then:

$$|x \cdot y| = |x \cdot (\lambda x)| = |\lambda| \|x\|^2 = \|x\| \|\lambda x\|.$$

If x and y are linearly independent, consider the vector $z = \lambda x + y$ for some $\lambda \in \mathbb{R}$. Then:

$$0 \leq \langle z, z \rangle = \|z\|^2 = \langle \lambda x + y, \lambda x + y \rangle .$$

Expanding this, we have:

$$\|z\|^2 = \lambda^2 \|x\|^2 + 2\lambda \langle x, y \rangle + \|y\|^2,$$

which is a quadratic function in λ. For $f(\lambda) \geq 0$ for all λ, the discriminant must be non-positive:

$$(2 \langle x, y \rangle)^2 - 4\|x\|^2 \|y\|^2 \leq 0.$$

Simplifying, we get:

$$4 \langle x, y \rangle^2 \leq 4\|x\|^2 \|y\|^2,$$

which leads to:

$$|x \cdot y| \leq \|x\| \|y\|.$$

□

1.7 Theorem: Inner Product and Norm

For any vectors $x, y \in \mathbb{R}^n$, the following relation holds:

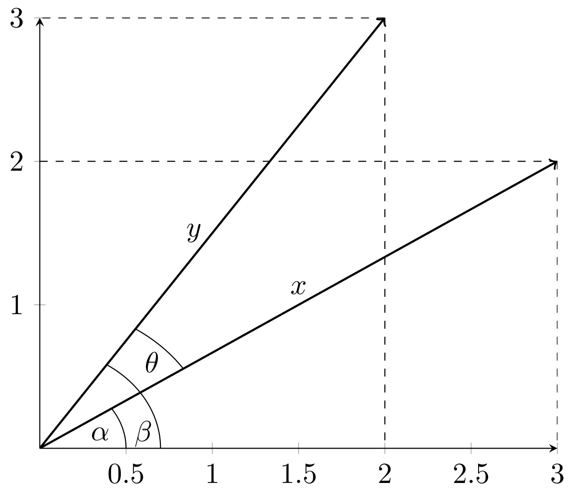
$$\langle x, y \rangle = \|x\| \|y\| \cos \theta,$$

where θ is the angle between the vectors x and y.

Proof. Consider the vectors x and y in $\mathbb{R}^2$. Let the angle between x and the positive x-axis be α, and the angle between y and the positive x-axis be β. Then, the angle between x and y is given by $\theta = \beta - \alpha$. We have:

$$x_1 = \|x\| \cos \alpha, \quad x_2 = \|x\| \sin \alpha,$$

$$y_1 = \|y\| \cos \beta, \quad y_2 = \|y\| \sin \beta.$$



Therefore, the inner product is:

$$\begin{aligned}
\langle x, y \rangle &= x_1 y_1 + x_2 y_2 \\
&= \|x\| \cos \alpha \cdot \|y\| \cos \beta + \|x\| \sin \alpha \cdot \|y\| \sin \beta \\
&= \|x\| \|y\| (\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\
&= \|x\| \|y\| \cos(\beta - \alpha) \\
&= \|x\| \|y\| \cos \theta.
\end{aligned}$$

□

Remark

From the theorem, we can derive the formula for the angle θ between two vectors x and y:

$$\theta = \arccos\left(\frac{\langle x, y\rangle}{\|x\|\|y\|}\right).$$

Note that if $x \perp y$ (i.e., x is perpendicular to y), then $\langle x, y\rangle = 0$ and thus $\theta = \frac{\pi}{2}$.

Examples

$C(a, b)$ is the set of continuous functions on the interval (a, b). For any $f, g \in C(a, b)$, we define the inner product as:

$$\langle f, g\rangle = \int_a^b f(x)g(x)\,dx.$$

We can add a weighted inner product as follows:

$$\langle f, g\rangle_w = \int_a^b f(x)g(x)w(x)\,dx,$$

where $w(x)$ is a weight function such as $w(x) = e^{-x}$ or $w(x) = 1/x^2$.
We might as well use the family of **orthogonal functions**:

$$\{\cos nx, \sin nx \mid n = 0, 1, 2, \ldots\} \quad \text{Fourier family.}$$

$$0 = \int_0^{2\pi} \cos nx \sin nx\,dx = \int_0^{2\pi} \cos nx \cos mx\,dx = \int_0^{2\pi} \sin nx \sin mx\,dx, \qquad n \neq m.$$

1.8 Vector Product in $\mathbb{R}^3$

Let $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ be two vectors in $\mathbb{R}^3$. The **vector product** (or cross product) of x and y is defined as:

$$x \times y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix}\mathbf{k},$$

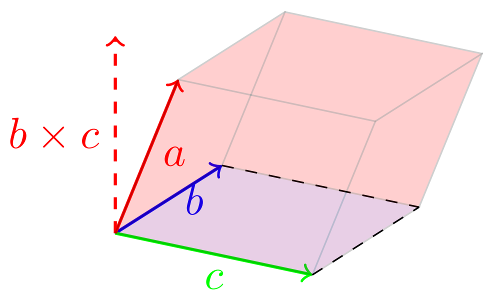
where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit vectors in $\mathbb{R}^3$.

1.8.1 Geometric Interpretation

To visualize the vector product, consider the triple product $a \cdot (b \times c)$, which represents the volume of the parallelepiped formed by the vectors a, b, and c:

$$a \cdot (b \times c) = (a_1, a_2, a_3) \cdot \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}\mathbf{k} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

If $a \in \text{span}\{b, c\}$, then the volume is zero, indicating that the vectors are coplanar.



Also,

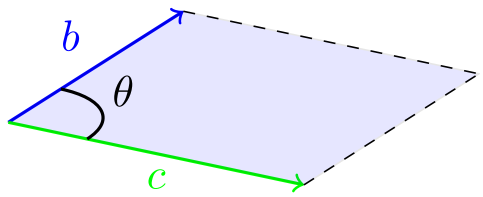
$$\|b \times c\|^2 = \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}^2 + \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}^2 + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}^2 = (b_2c_3 - b_3c_2)^2 + (b_1c_3 - b_3c_1)^2 + (b_1c_2 - b_2c_1)^2 =$$

$$= (b_1^2 + b_2^2 + b_3^2)(c_1^2 + c_2^2 + c_3^2) - (b_1c_1 + b_2c_2 + b_3c_3)^2 = \|b\|^2\|c\|^2 - (b \cdot c)^2 = \|b\|^2\|c\|^2 \sin^2\theta.$$

Thus, we have:

$$\|b \times c\| = \|b\|\|c\| \sin\theta,$$

where θ is the angle between vectors b and c. This represents the area of the parallelogram formed by b and c.



2 Limits, Continuity, and Differentiability

2.1 Level Curves

Let $A \subset \mathbb{R}^n$ be a subset of $\mathbb{R}^n$ and let $f : A \to \mathbb{R}$ be a real-valued function defined on A. For any constant $c \in \mathbb{R}$, the **level set** of f corresponding to the value c is defined as:

$$L_c = \{x \in A \mid f(x) = c\}.$$

In the case where $n = 2$, the level sets are called **level curves** or **contour lines**. In the case where $n = 3$, the level sets are called **level surfaces**.

Example (Chapter 12, Ex 3)

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x, y) = xy.$$

The level curves of f for different values of c are given by:

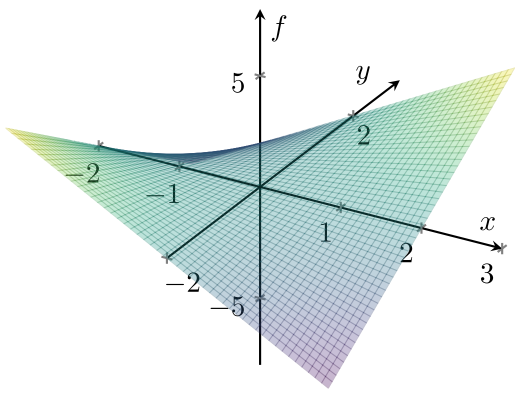
- For $c = -1$: $L_{-1} = \{(x, y) \in \mathbb{R}^2 \mid xy = -1\}$.
- For $c = 1$: $L_1 = \{(x, y) \in \mathbb{R}^2 \mid xy = 1\}$.
- For $c = 3$: $L_3 = \{(x, y) \in \mathbb{R}^2 \mid xy = 3\}$.

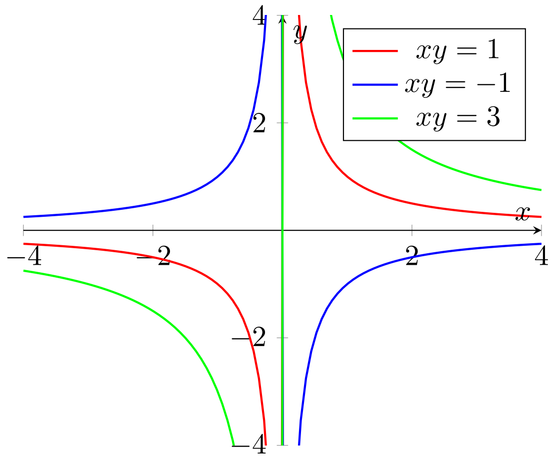
2.1.1 Graph of a Function

The **graph** of a function $f : A \to \mathbb{R}$ is defined as the set of points in $\mathbb{R}^{n+1}$ given by:

$$G_f = \{(x, f(x)) \mid x \in A\}.$$

If for instance, $f : \mathbb{R}^2 \to \mathbb{R}$, then the graph of f is a surface in $\mathbb{R}^3$.





Remark

Level sets $F : U \subset \mathbb{R}^n \to \mathbb{R}^m$; U open set:

$$F^{-1}(c) = \{x \in U \mid F(x) = c\}, \quad c = (c_1, c_2, \ldots, c_m)$$

In terms of the coordinate functions $F = (f_1, f_2, \ldots, f_m)$, we have:

$$F(x)(c), \text{ for } x \in \mathbb{R}^n \implies f_i(x_1, x_2, \ldots, x_n) = c_i, \quad i = 1, 2, \ldots, m.$$

Hence,

$$F^{-1}(c) = \bigcap_{i=1}^{m} \{x \in \mathbb{R}^n \mid f_i(x) = c_i\}.$$

Example

Let $F : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by:

$$F(x, y, z) = (x^2 + y^2 + z^2 - 1, 2x^2 + 2y^2 - 1).$$

The coordinate functions are:

$$\begin{aligned} f_1^{-1}(0) &= \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 - 1 = 0\}, \\ f_2^{-1}(0) &= \{(x, y, z) \in \mathbb{R}^3 \mid 2x^2 + 2y^2 - 1 = 0\}. \end{aligned}$$

To find the level set we solve the system of equations:

$$\begin{aligned} x^2 + y^2 + z^2 - 1 &= 0, \\ 2x^2 + 2y^2 - 1 &= 0. \end{aligned}$$

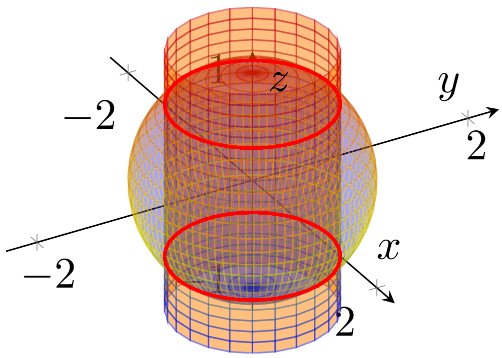
From the second equation, we have:

$$x^2 + y^2 = \frac{1}{2}.$$

Substituting this into the first equation gives:

$$\frac{1}{2} + z^2 - 1 = 0 \implies z^2 = \frac{1}{2} \implies z = \pm\frac{1}{\sqrt{2}}.$$

Thus, the level set $F^{-1}(0)$ is the intersection of the sphere $x^2 + y^2 + z^2 = 1$ and the cylinder $2x^2 + 2y^2 = 1$.



Remarks

- There is a connection between level sets and graphs. Firstly, the graph of a function $f : A \subset \mathbb{R}^n \to \mathbb{R}$ can be viewed as the level set of the function $F : A \times \mathbb{R} \to \mathbb{R}$ defined by:

$$F : U \subset \mathbb{R}^n \to \mathbb{R}^m, \quad F(x, z) = f(x) - z.$$

- The level curves allow us to draw tri-dimensional surfaces on a two-dimensional plane.

- The distance between two level curves (they can never intersect) gives us information about the steepness of the surface. If the level curves are close together, the surface is steep; if they are far apart, the surface is flat.

2.2 Limits of Functions

Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a function, and let $a \in \mathbb{R}^n$ be a point in its domain. We say that the limit of $f(x)$ as x approaches a is $L \in \mathbb{R}^m$, denoted by:

$$\lim_{x \to a} f(x) = L,$$

if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in \mathbb{R}^n$ satisfying $0 < \|x - a\| < \delta$, we have:

$$\|f(x) - L\| < \epsilon.$$

2.2.1 Methods to Compute Limits

1. **Applying the Definition:** We must prove that for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$0 < (x, y) - (a, b) < \delta \implies |f(x, y) - L| < \epsilon.$$

2. **Iterative limits:**

$$\lim_{x \to a} \left(\lim_{y \to b} f(x, y) \right) \neq \lim_{y \to b} \left(\lim_{x \to a} f(x, y) \right).$$

This gives non-existence of the limit, so we cannot prove existence by this method.

3. **Following families of functions:** If we can find two different paths approaching the same point that yield different limit values, then the limit does not exist.

2.3 Higher Order Derivatives

Partial derivatives of first order $f(x_1, x_2, \ldots, x_n)$

$$\frac{\partial f}{\partial x_1}, \quad \frac{\partial f}{\partial x_2}, \quad \ldots, \quad \frac{\partial f}{\partial x_n}.$$

Second order partial derivatives:

$$\frac{\partial^2 f}{\partial x_1^2}, \quad \frac{\partial^2 f}{\partial x_i \partial x_j}, \quad \ldots, \quad \frac{\partial^2 f}{\partial x_n^2}.$$

In general, the k-th order partial derivatives of f are denoted by:

$$\frac{\partial^k f}{\partial x_{i_1} \partial x_{i_2} \ldots \partial x_{i_k}}, \quad \text{where } x_{i_j} \neq x_{i_k}.$$

Remark

We can say that the function f is of class C^k if all its partial derivatives up to order k exist and are continuous. Also, a function is **analytic** if it is of class C^∞.

Example

Consider the function $f : \mathbb{R}^3 \to \mathbb{R}$ defined by:

$$f(x, y, z) = e^{xy} + z \cos x.$$

The first order partial derivatives are:

$$\frac{\partial f}{\partial x} = ye^{xy} - z \sin x, \quad \frac{\partial f}{\partial y} = xe^{xy}, \quad \frac{\partial f}{\partial z} = \cos x.$$

The second order partial derivatives are:

$$\frac{\partial^2 f}{\partial x^2} = y^2 e^{xy} - z \cos x, \quad \frac{\partial^2 f}{\partial y \partial x} = e^{xy} + xye^{xy}, \quad \frac{\partial^2 f}{\partial y^2} = x^2 e^{xy}, \quad \frac{\partial^2 f}{\partial z^2} = 0 \ldots$$

Remark

We call $\dfrac{\partial^k f}{\partial x_{i_1} \partial x_{i_2} \dots \partial x_{i_k}}$ a **mixed or crossing partial derivative** if at least two of the indices i_j are different. For example, $\dfrac{\partial^2 f}{\partial x \partial y}$ is a mixed partial derivative, while $\dfrac{\partial^2 f}{\partial x^2}$ is not.

2.3.1 Schwarz's Theorem

Take a function $f : A \subset \mathbb{R}^2 \to \mathbb{R}$ of class C^1. Then, the mixed partial derivatives of f are equal:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Remark

In general, if $f : A \subset \mathbb{R}^n \to \mathbb{R}$ is of class C^{k-1}, then the order of differentiation does not matter for any mixed partial derivatives of order less than or equal to k. That is, for any indices $i_1, i_2, \dots, i_k$ and any permutation σ of $\{1, 2, \dots, k\}$, we have:

$$\frac{\partial^k f}{\partial x_{i_1} \partial x_{i_2} \dots \partial x_{i_k}} = \frac{\partial^k f}{\partial x_{i_{\sigma(1)}} \partial x_{i_{\sigma(2)}} \dots \partial x_{i_{\sigma(k)}}}.$$

Proof. To simplify the proof, we assume that the derivatives are taken at the point $(0, 0)$. Take an arbitrary point (x, y) inside a ball centered at $(0, 0)$:

$$\|(x, y)\| < \varepsilon, \quad \text{assuming } B_\varepsilon(0, 0).$$

Then, we can **discretize** the derivatives as follows:

$$\frac{\partial f(x, y)}{\partial x} \approx \frac{f(x + h, y) - f(x - h, y)}{2h},$$

$$\frac{\partial f(x, y)}{\partial y} \approx \frac{f(x, y + k) - f(x, y - k)}{2k}.$$

And for the second order derivatives:

$$\frac{\partial^2 f(x, y)}{\partial x^2} \approx \frac{f(x + h, y) - 2f(x, y) + f(x - h, y)}{h^2},$$

$$\frac{\partial^2 f(x, y)}{\partial y^2} \approx \frac{f(x, y + k) - 2f(x, y) + f(x, y - k)}{k^2}.$$

For the mixed partial derivative, we have:

$$\frac{\partial^2 f(x, y)}{\partial x \partial y} \approx \frac{f(x + h, y + k) - f(x + h, y - k) - f(x - h, y + k) + f(x - h, y - k)}{4hk}.$$

Back to the proof, we use the second differentiation:

$$D(x, y) = f(x, y) - f(x, 0) - f(0, y) + f(0, 0).$$

Now, we define the function $\varphi : \mathbb{R} \to \mathbb{R}$ by:

$$\left.\begin{aligned} \varphi(x) &= f(x, y) - f(x, 0), \\ \varphi(0) &= f(0, y) - f(0, 0). \end{aligned}\right\} \implies D(x, y) = \varphi(x) - \varphi(0).$$

Then, we have:
$$\varphi'(x) = \frac{\partial f(x,y)}{\partial x} - \frac{\partial f(x,0)}{\partial x},$$
and by the Mean Value Theorem, there exists $-x \leq \hat{x} \leq x$ such that:
$$D(x,y) = \varphi(x) - \varphi(0) = x\varphi'(\hat{x}) = x\left(\frac{\partial f(\hat{x},y)}{\partial x} - \frac{\partial f(\hat{x},0)}{\partial x}\right).$$
Applying the Mean Value Theorem again, now for the map:
$$y \longmapsto \frac{\partial f(\hat{x},y)}{\partial x},$$
we find that
$$D(x,y) = xy\frac{\partial^2 f(\hat{x},\hat{y})}{\partial y \partial x}, \quad \text{for some } -y \leq \hat{y} \leq y$$
.
Similarly, we define the function $\psi : \mathbb{R} \to \mathbb{R}$ by:
$$\left.\begin{array}{l} \psi(y) = f(x,y) - f(0,y), \\ \psi(0) = f(x,0) - f(0,0). \end{array}\right\} \implies D(x,y) = \psi(y) - \psi(0).$$
By following the same steps, we find that:
$$D(x,y) = xy\frac{\partial^2 f(\tilde{x},\tilde{y})}{\partial x \partial y}, \quad \text{for some } -x \leq \tilde{x} \leq x, -y \leq \tilde{y} \leq y.$$
Since f is of class C^1, we have:
$$\frac{\partial^2 f(\hat{x},\hat{y})}{\partial y \partial x} = \frac{\partial^2 f(\tilde{x},\tilde{y})}{\partial x \partial y}.$$
Passing to the limit as $(x,y) \to (0,0)$, we conclude that:
$$\frac{\partial^2 f(0,0)}{\partial y \partial x} = \frac{\partial^2 f(0,0)}{\partial x \partial y}.$$
□

3 Local Extrema

Let $f : A \subset \mathbb{R}^n \to \mathbb{R}$ be a function, and let $a \in A$ be a point in its domain. We say that f has a **local maximum** at a if there exists a neighborhood U of a such that for all $x \in U$, we have:
$$f(x) \leq f(a).$$
Similarly, f has a **local minimum** at a if there exists a neighborhood U of a such that for all $x \in U$, we have:
$$f(x) \geq f(a).$$

Definition

Given a function $f : A \subset \mathbb{R}^n \to \mathbb{R}$ and a point $a \in A$, we say that a is a **critical point** or **stationary point** of f if the gradient of f at a is zero, i.e.,
$$\nabla f(a) = 0.$$

Remark

A critical point that is not a local extremum is called a **saddle point**.

- **Theorem:** If $f : A \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable and has a local extremum at a, then a is a critical point of f.
- If a is a local extremum, and the directional derivative of f at a in any direction v exists, then the directional derivative must be zero:

$$D_v f(a) = \nabla f(a) \cdot v = 0.$$

- If f is differentiable, and we find that $\nabla f(a) = 0$, then every directional derivative at a is zero.
- If any of the directional derivatives at a does not exist, then the function is not differentiable at a, and a is an **angular point**.
- In both cases we can find a potential local extremum at a, but it can also be a saddle point.

3.1 How to find local extrema?

Let's consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x, y) = 3x^2 + y^2 - 6x - 4y + 8.$$

To find the local extrema of f, we first compute the gradient:

$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (6x - 6, 2y - 4).$$

Setting the gradient equal to zero gives us the critical point:

$$6x - 6 = 0 \implies x = 1, \quad 2y - 4 = 0 \implies y = 2.$$

Thus, the critical point is $(1, 2)$. We have:

$$f(1, 2) = 3(1)^2 + (2)^2 - 6(1) - 4(2) + 8 = 3 + 4 - 6 - 8 + 8 = 1.$$

We will complete the squares to determine the nature of the critical point:

$$\begin{aligned} f(x, y) &= 3x^2 - 6x + y^2 - 4y + 8 \\ &= 3(x^2 - 2x) + (y^2 - 4y) + 8 \\ &= 3(x - 1)^2 - 3 + (y - 2)^2 - 4 + 8 \\ &= 3(x - 1)^2 + (y - 2)^2 + 1. \end{aligned}$$

Since $f(x, y) \geq 1 = f(1, 2)$ for all $(x, y) \in \mathbb{R}^2$, the critical point $(1, 2)$ is a local minimum of f.

Example

Find the extrema of the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x, y) = 1 - \sqrt{x^2 + y^2}.$$

We are looking for the points where:

- The partial derivatives are zero (stationary points), or
- The function is not differentiable (angular points).

First, we compute the partial derivatives:

$$\frac{\partial f}{\partial x} = -\frac{x}{\sqrt{x^2 + y^2}}, \quad \frac{\partial f}{\partial y} = -\frac{y}{\sqrt{x^2 + y^2}}.$$

Setting the partial derivatives equal to zero gives us:

$$-\frac{x}{\sqrt{x^2 + y^2}} = 0 \implies x = 0, \quad -\frac{y}{\sqrt{x^2 + y^2}} = 0 \implies y = 0.$$

Thus, the function f is not differentiable at $(0, 0)$ since the partial derivatives are undefined at that point. However, we can still evaluate f at $(0, 0)$:

$$f(0, 0) = 1 - \sqrt{0^2 + 0^2} = 1.$$

For any other point $(x, y) \neq (0, 0)$, we have:

$$f(x, y) = 1 - \sqrt{x^2 + y^2} < 1.$$

Thus, the point $(0, 0)$ is a global maximum of f.

Example

Another example is the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x, y) = 1 - x^2 + y^2.$$

We compute the partial derivatives:

$$\left.\begin{aligned} \tfrac{\partial f}{\partial x} &= -2x \\ \tfrac{\partial f}{\partial y} &= 2y. \end{aligned}\right\} \implies (0, 0) \text{ is a stationary point.}$$

Evaluating f at $(0, 0)$ gives us:

$$f(0, 0) = 1 - 0^2 - 0^2 = 1.$$

For any other point $(x, y) \neq (0, 0)$, we have:

$$f\big|_{x=0}(x, y) = f(0, y) = 1 + y^2 > 1, \quad f\big|_{y=0}(x, y) = f(x, 0) = 1 - x^2 < 1.$$

Thus, the point $(0, 0)$ is a saddle point of f.

3.2 Methods for Classification of Extrema

To classify the extrema systematically, we will use:

- Quadratic forms
- The Hessian matrix.

3.2.1 Second Derivative (general definition)

Take a vector $v = (v_1, \dots, v_n) \in \mathbb{R}^n$ and a function $f : A \subset \mathbb{R}^n \to \mathbb{R}$, differentiable at a. Then,

$$D_v f(a) = \langle \nabla f(a), v \rangle = \sum_{i=1}^{n} v_i \cdot \frac{\partial f}{\partial x_i}(a).$$

Then, we can define the second derivative of f at a in the direction of v as:

$$D_v^2 f(a) = D_v(D_v f)(a) = \sum_{j=1}^{n} v_j \cdot \frac{\partial}{\partial x_j} \left(\sum_{i=1}^{n} v_i \cdot \frac{\partial f}{\partial x_i}(a) \right) = \sum_{i=1}^{n} \sum_{j=1}^{n} v_i v_j \cdot \frac{\partial^2 f}{\partial x_i \partial x_j}(a) = v^T H_f(a) v,$$

where $H_f(a)$ is the Hessian matrix of f at a.

$$\boxed{D_v^2 f(a) = v^T H_f(a) v}$$

3.2.2 Hessian Matrix

The **Hessian matrix** of a function $f : A \subset \mathbb{R}^n \to \mathbb{R}$ at a point $a \in A$ is the $n \times n$ matrix given by:

$$H_f(a) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(a) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(a) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(a) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(a) & \frac{\partial^2 f}{\partial x_2^2}(a) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(a) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(a) & \frac{\partial^2 f}{\partial x_n \partial x_2}(a) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(a) \end{bmatrix}.$$

Example

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x, y) = \log(x^2 + y^3)$$

The partial derivatives are:

$$\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^3}, \quad \frac{\partial f}{\partial y} = \frac{3y^2}{x^2 + y^3}.$$

The second order partial derivatives are:

$$\frac{\partial^2 f}{\partial x^2} = \frac{2(y^3 - x^2)}{(x^2 + y^3)^2}, \quad \frac{\partial^2 f}{\partial y^2} = \frac{6yx^2 - 3y^4}{(x^2 + y^3)^2}, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{-6xy^2}{(x^2 + y^3)^2}.$$

Thus, the Hessian matrix of f at (x, y) is:

$$H_f(x, y) = \begin{bmatrix} \frac{2(y^3 - x^2)}{(x^2 + y^3)^2} & \frac{-6xy^2}{(x^2 + y^3)^2} \\ \frac{-6xy^2}{(x^2 + y^3)^2} & \frac{6yx^2 - 3y^4}{(x^2 + y^3)^2} \end{bmatrix}.$$

3.2.3 Quadratic Forms

A **quadratic form** on $\mathbb{R}^n$ is a function $Q : \mathbb{R}^n \to \mathbb{R}$ defined by:

$$Q(x) = \sum_{i,j=1}^{n} a_{ij}x_ix_j = X^TAX = a_{11}x_1^2 + a_{22}x_2^2 + \cdots + a_{nn}x_n^2 + \sum_{\substack{1\le i,j\le n \\ i\ne j}} a_{ij}x_ix_j,$$

where a_{ij} are constants. For example, the function $Q_1 : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$Q_1(x_1, x_2) = x_1^2 - x_2^2$$

is a quadratic form on $\mathbb{R}^2$, and so is the function $Q_2 : \mathbb{R}^3 \to \mathbb{R}$ defined by:

$$Q_2(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 5x_3^2 + 2x_1x_2 - 2x_1x_3 + 2x_2x_3.$$

Quadratic forms can be classified as follows:

- A quadratic form Q is **positive definite** if $Q(x) > 0$ for all $x \in \mathbb{R}^n$.
- A quadratic form Q is **negative definite** if $Q(x) < 0$ for all $x \in \mathbb{R}^n$.
- A quadratic form Q is **indefinite** if $\exists\ x, y \in \mathbb{R}^n$ such that $Q(x) > 0$ and $Q(y) < 0$.

Any quadratic form Q can be associated with a symmetric matrix A such that $Q(x) = x^TAx$. For example, the quadratic form $Q_2(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 5x_3^2 + 2x_1x_2 - 2x_1x_3 + 2x_2x_3$ can be represented by the symmetric matrix:

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 5 \end{bmatrix}.$$

Examples

Consider the quadratic form $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ defined by:

$$\|x\|^2 = x^TIx = \sum_{i=1}^{n} x_i^2,$$

where I is the identity matrix. This is a positive definite quadratic form.

The quadratic form $Q_1 : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$Q(x_1, x_2) = x_1^2 - x_2^2$$

is an indefinite quadratic form, since $Q(1, 0) = 1 > 0$ and $Q(0, 1) = -1 < 0$.

The quadratic form $Q_2 : \mathbb{R}^3 \to \mathbb{R}$ defined by:

$$Q_2(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 5x_3^2 + 2x_1x_2 - 2x_1x_3 + 2x_2x_3$$

is a semi-positive definite quadratic form, since it can be rewritten as:

$$Q_2(x_1, x_2, x_3) = (x_1 + x_2 - x_3)^2 + (x_2 + 2x_3)^2,$$

which is a sum of squares and hence always non-negative.

Proposition

A quadratic form $Q : \mathbb{R}^n \to \mathbb{R}$ can be expressed as a sum of squares of linear forms. That is, there exist linear forms $L_1, L_2, \ldots, L_m : \mathbb{R}^n \to \mathbb{R}$ such that:

$$Q(x_1, x_2, \ldots, x_n) = L_1^2 \pm L_2^2 \pm \cdots \pm L_n^2,$$

where $L_i, \ldots, L_n$ do not depend on the variables $x_1, x_2, \ldots, x_{i-1}$ for $i = 1, 2, \ldots, n$.

Example

Consider the quadratic form $Q : \mathbb{R}^3 \to \mathbb{R}$ defined by:

$$Q(x_1, x_2, x_3) = x_1^2 - x_2^2 + x_3^2 + 2x_1x_2 - 4x_1x_3.$$

We can rewrite this as:

$$\begin{aligned} Q(x_1, x_2, x_3) &= (x_1 + x_2 - 2x_3)^2 - 2x_2^2 - 2x_3^2 \\ &= (x_1 + x_2 - 2x_3)^2 - 2x_2^2 + 4x_2x_3 - 3x_3^2 \\ &= (x_1 + x_2 - 2x_3)^2 - 2(x_2 - x_3)^2 - x_3^2. \end{aligned}$$

Remark

If we do not have squares in the expression of a quadratic form, we can always complete the squares to rewrite it as a sum of squares of linear forms. For example, the quadratic form $Q : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$Q(x_1, x_2) = x_1x_2 + 2x_2x_3 \begin{cases} x_1 = y_1 - y_2, \\ x_2 = y_1 + y_2 \\ x_3 = y_3 \end{cases} \implies$$

$$\implies Q(y_1, y_2, y_3) = y_1^2 - y_2^2 + 2y_1y_3 + 2y_2y_3$$

Remark

Given X^TAX, we can express $A = PDP^{-1}$ (A symmetric) and with all the eigenvalues of A in D being real,

$$A = PDP^{-1}, \quad D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

where λ_i are the eigenvalues of A. If we now perform the change of variables $X = PY$, we have:

$$Q(X) = X^TAX = (PY)^TA(PY) = Y^TP^TAPY = Y^TDY = \lambda_1y_1^2 + \lambda_2y_2^2 + \cdots + \lambda_ny_n^2.$$

Theorem

Let $Q : \mathbb{R}^n \to \mathbb{R}$ be a quadratic form. Then:

- Q is positive definite if and only if all the eigenvalues of the associated matrix A are positive.
- Q is negative definite if and only if all the eigenvalues of the associated matrix A are negative.
- Q is indefinite if and only if there exist both positive and negative eigenvalues of the associated matrix A.

3.2.4 Taylor's Polynomial

Let $f : A \subset \mathbb{R}^2 \to \mathbb{R}$ be a differentiable function, and let $a = (x_0, y_0) \in A$. The **Taylor's polynomial** of f at a is the polynomial defined by:

$$f(x,y) = f(x_0,y_0) + \nabla f(x_0,y_0) \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} + \frac{1}{2}(x - x_0, y - y_0) \cdot H_f(x_0,y_0) \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} + R,$$

where $H_f(x_0, y_0)$ is the Hessian matrix of f at a.
One can see that it is composed of three terms:

- The first term $f(x_0, y_0)$ is the value of the function at the point a.
- The second term $\nabla f(x_0,y_0) \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$ is the linear approximation of f at a.
- The third term $\frac{1}{2}(x - x_0, y - y_0) \cdot H_f(x_0,y_0) \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$ is the quadratic approximation of f at a.

3.3 Classification of Extrema

Let $U \subset \mathbb{R}^n$ be an open set, and let $f : U \to \mathbb{R}$ be a function of class C^1. Let $a \in U$ be a critical point of f. If

$$\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \frac{\partial f}{\partial x_2}(a), \ldots, \frac{\partial f}{\partial x_n}(a) \right) = 0,$$

then

$$f(x) - f(a) = \frac{1}{2}(x - a)^T H_f(a)(x - a),$$

where $H_f(a)$ is the Hessian matrix of f at a. Thus, the nature of the critical point a can be determined by the definiteness of the quadratic form associated with the Hessian matrix:

- If the quadratic form $Q(x) = (x - a)^T H_f(a)(x - a)$ is positive definite, then a is a local minimum of f.
- If the quadratic form $Q(x) = (x - a)^T H_f(a)(x - a)$ is negative definite, then a is a local maximum of f.
- If the quadratic form $Q(x) = (x - a)^T H_f(a)(x - a)$ is indefinite, then a is a saddle point of f.

3.3.1 Hessian or Sylvester's Criteria in $\mathbb{R}^2$

Given the Hessian matrix of a function $f : A \subset \mathbb{R}^2 \to \mathbb{R}$ at a critical point $a = (x, y)$:

$$H_f(a) = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x^2}(a) & \dfrac{\partial^2 f}{\partial x \partial y}(a) \\ \dfrac{\partial^2 f}{\partial y \partial x}(a) & \dfrac{\partial^2 f}{\partial y^2}(a) \end{bmatrix},$$

the determinant of the Hessian matrix is given by:

$$\det(H_f(a)) = \frac{\partial^2 f}{\partial x^2}(a) \cdot \frac{\partial^2 f}{\partial y^2}(a) - \left(\frac{\partial^2 f}{\partial x \partial y}(a) \right)^2.$$

Then, the classification of the critical point $a = (x_0, y_0)$ can be determined as follows:

- If $\det(H_f(x_0, y_0)) > 0$ and $\dfrac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$, then a is a local minimum of f.
- If $\det(H_f(x_0, y_0)) > 0$ and $\dfrac{\partial^2 f}{\partial x^2}(x_0, y_0) < 0$, then a is a local maximum of f.
- If $\det(H_f(x_0, y_0)) < 0$, then a is a saddle point of f.
- If $\det(H_f(x_0, y_0)) = 0$, then the test is inconclusive, and we need to use higher-order derivatives or other methods to classify the critical point.

Example

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x, y) = x^2 + 2y^2 - 4y.$$

The gradient of f is:

$$\nabla f(x, y) = (2x, 4y - 4).$$

Setting the gradient equal to zero gives us the critical point:

$$2x = 0 \implies x = 0, \quad 4y - 4 = 0 \implies y = 1.$$

Thus, the critical point is $(0, 1)$. The Hessian matrix of f at $(0, 1)$ is:

$$H_f(0, 1) = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}.$$

The determinant of the Hessian matrix is:

$$\det(H_f(0, 1)) = (2)(4) - (0)^2 = 8 > 0,$$

and $\dfrac{\partial^2 f}{\partial x^2}(0, 1) = 2 > 0$. Therefore, the critical point $(0, 1)$ is a local minimum of f.

Example

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x, y) = 8x^3 - 24xy + 3y^2.$$

The gradient of f is:

$$\nabla f(x, y) = (24x^2 - 24y, -24x + 3y^2).$$

Setting the gradient equal to zero gives us the critical points:

$$\begin{aligned} x^2 - y = 0 &\implies y = x^2, \\ -8x + y^2 = 0 &\implies y^2 = 8x. \end{aligned}$$

Substituting $y = x^2$ into $y^2 = 8x$ gives us:

$$x^4 = 8x \implies x^4 - 8x = 0 \implies x(x^3 - 8) = 0 \implies x = 0 \text{ or } x^3 = 8 \implies x = 2.$$

Thus, the critical points are $(0, 0)$ and $(2, 4)$. The Hessian matrices of f at $(0, 0)$ and $(2, 4)$ are:

$$H_f(0, 0) = \begin{bmatrix} 0 & -24 \\ -24 & 0 \end{bmatrix}, \quad H_f(2, 4) = \begin{bmatrix} 96 & -24 \\ -24 & 24 \end{bmatrix}.$$

The determinants of the Hessian matrix at said points are:

$$\det(H_f(0,0)) = (0)(0) - (-24)^2 = -576 < 0,$$
$$\det(H_f(2,4)) = (96)(24) - (-24)^2 = 2304 - 576 = 1728 > 0.$$

Since $\det(H_f(0,0)) < 0$, the critical point $(0,0)$ is a saddle point of f. Since $\det(H_f(2,4)) > 0$ and $\frac{\partial^2 f}{\partial x^2}(2,4) = 96 > 0$, the critical point $(2,4)$ is a local minimum of f.

3.3.2 Extension to $\mathbb{R}^3$

The hessian matrix of a function $f : A \subset \mathbb{R}^3 \to \mathbb{R}$ is of the form:

$$H_f = \begin{bmatrix} A & B & C \\ D & E & F \\ G & H & I \end{bmatrix} \approx \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}.$$

Looking at the determinants of the leading principal minors of H_f, we have:

- $\Delta_1 = A$
- $\Delta_2 = \det \begin{pmatrix} A & B \\ D & E \end{pmatrix} = AE - BD$
- $\Delta_3 = \det(H_f) = A(EI - FH) - B(DI - FG) + C(DH - EG)$

With the eigenvalues of H_f being $\lambda_1, \lambda_2, \lambda_3$, we have:

- If $\Delta_1 > 0$, $\Delta_2 > 0$, and $\Delta_3 > 0$, then f is positive definite, and the critical point is a local minimum.
- If $\Delta_1 < 0$, $\Delta_2 > 0$, and $\Delta_3 < 0$, then f is negative definite, and the critical point is a local maximum.
- Any other combination of signs for Δ_1, Δ_2, and Δ_3 indicates that the critical point is a saddle point.